

Deepika Dandolu

Exercise 2

**Pneumonia
Classification**

Results (SVM linear)

Chest X-Ray Class Counts - 20-80 Split

	Index	Chest X-Ray Class	Count
0	1	NORMAL	268
1	2	PNEUMONIA	775

Chest X-Ray Class Counts - 40-60 Split

	Index	Chest X-Ray Class	Count
0	1	NORMAL	536
1	2	PNEUMONIA	1550

Chest X-Ray Class Counts - 60-40 Split

	Index	Chest X-Ray Class	Count
0	1	NORMAL	804
1	2	PNEUMONIA	2325

Chest X-Ray Class Counts - 80-20 Split

	Index	Chest X-Ray Class	Count
0	1	NORMAL	1073
1	2	PNEUMONIA	3099

Original shape: (4172, 40000)
PCA reduced shape: (4172, 1523)

Linear SVM Results

20-80 Split

Training Score: 1.0

Testing Score: 0.9482758620689655

- Training Data Shape : (2086, 40000)
Testing Data Shape : (3130, 40000)
PCA Reduced Training Shape : (2086, 976)
PCA Reduced Testing Shape : (3130, 976)

40-60 Split

Linear SVM Results

Training Score : 1.0

Testing Score : 0.9539936102236422

Training Data Shape : (3129, 40000)
Testing Data Shape : (2087, 40000)
PCA Reduced Training Shape : (3129, 1277)
PCA Reduced Testing Shape : (2087, 1277)

60-40 Split

Linear SVM Results

Training Score : 1.0

Testing Score : 0.9535218016291327

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Training Size : 20%
Training Shape: (1043, 40000)
Testing Shape : (4173, 40000)
PCA Training Shape: (1043, 594)
PCA Testing Shape : (4173, 594)

Linear SVM Results (20-80 Split)
Accuracy : 0.9513539420081476
F1-Score : 0.950780466046371
Sensitivity: 0.9252104440369179
Specificity: 0.9252104440369179

Training Size : 40%
Training Shape: (2086, 40000)
Testing Shape : (3130, 40000)
PCA Training Shape: (2086, 976)
PCA Testing Shape : (3130, 976)

Linear SVM Results (40-60 Split)
Accuracy : 0.9539936102236422
F1-Score : 0.9537042970779374
Sensitivity: 0.9337046684031256
Specificity: 0.9337046684031256

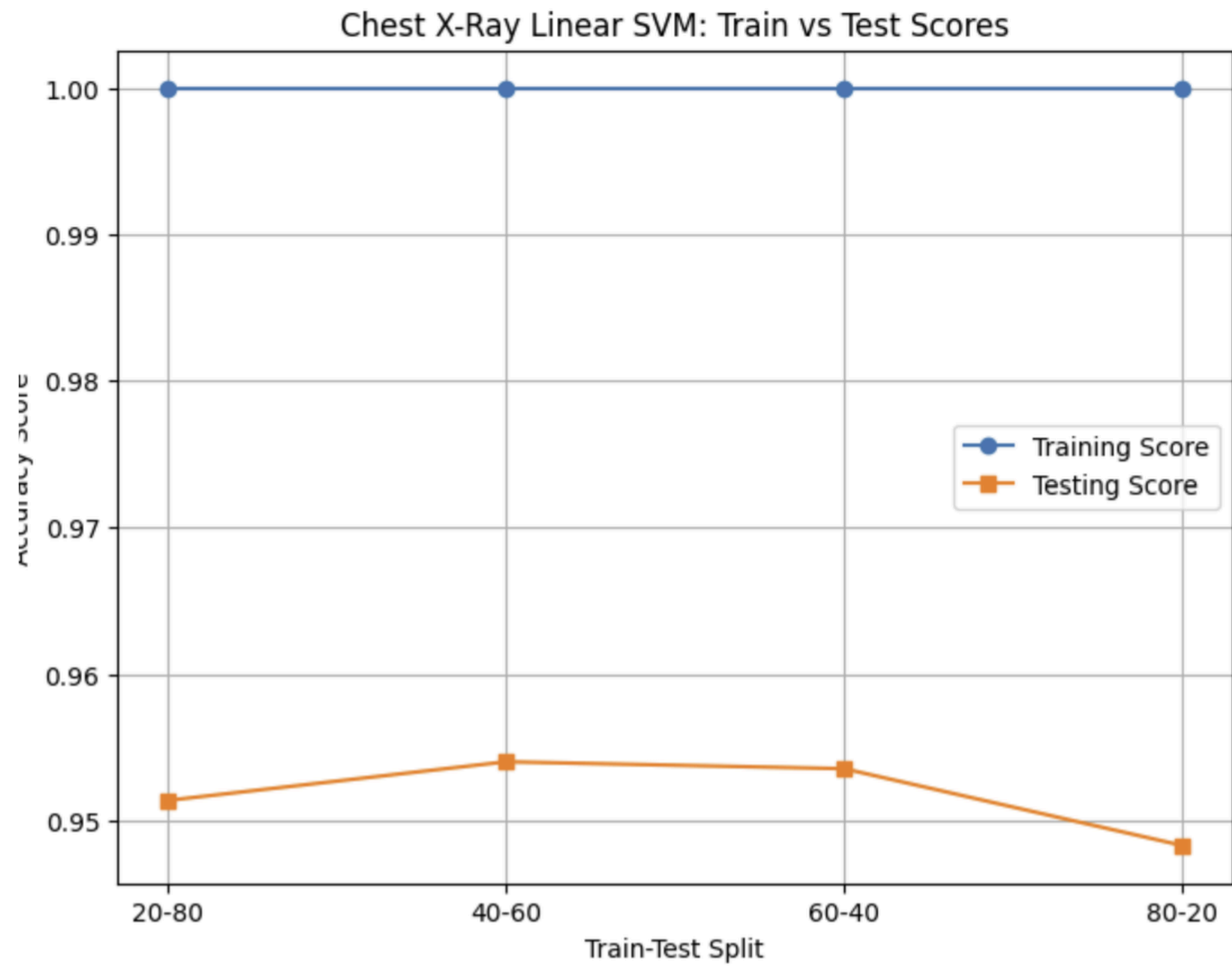
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Training Size : 60%
Training Shape: (3129, 40000)
Testing Shape : (2087, 40000)
PCA Training Shape: (3129, 1277)
PCA Testing Shape : (2087, 1277)

Linear SVM Results (60-40 Split)
Accuracy : 0.9535218016291327
F1-Score : 0.9532747864499224
Sensitivity: 0.9340241484952243
Specificity: 0.9340241484952243

Training Size : 80%
Training Shape: (4172, 40000)
Testing Shape : (1044, 40000)
PCA Training Shape: (4172, 1523)
PCA Testing Shape : (1044, 1523)

Linear SVM Results (80-19 Split)
Accuracy : 0.9482758620689655
F1-Score : 0.9478820311442842
Sensitivity: 0.924901907985844
Specificity: 0.924901907985844



RESULTS (CNN)

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Chest X-Ray Class Counts - 20-80 Split

	Index	Chest X-Ray Class	Count
0	1	NORMAL	268
1	2	PNEUMONIA	775

Chest X-Ray Class Counts - 40-60 Split

	Index	Chest X-Ray Class	Count
0	1	NORMAL	536
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Chest X-Ray Class Counts - 60-40 Split

Chest X-Ray Class Counts - 60-40 Split

	Index	Chest X-Ray Class	Count
0	1	NORMAL	804
1	2	PNEUMONIA	2325

Chest X-Ray Class Counts - 80-20 Split

	Index	Chest X-Ray Class	Count
0	1	NORMAL	1073
1	2	PNEUMONIA	3099

Training Size : 20%
Training Shape: (1043, 200, 200, 1)
Testing Shape : (4173, 200, 200, 1)
/usr/local/lib/python3.12/dist-packa
super().__init__(activity_regulariz

CNN Results (20-80 Split)
Accuracy : 0.9657320872274143
F1-Score : 0.9657992544999009
Sensitivity: 0.9571304452394552
Specificity: 0.9571304452394552

Training Size : 60%
Training Shape: (3129, 200, 200, 1)
Testing Shape : (2087, 200, 200, 1)
/usr/local/lib/python3.12/dist-packag
super().__init__(activity_regulariz

CNN Results (60-40 Split)
Accuracy : 0.9693339722089123
F1-Score : 0.9693339722089123
Sensitivity: 0.9598822610680603
Specificity: 0.9598822610680603

Training Size : 40%
Training Shape: (2086, 200, 200, 1)
Testing Shape : (3130, 200, 200, 1)
/usr/local/lib/python3.12/dist-packa
super().__init__(activity_regulariz

CNN Results (40-60 Split)
Accuracy : 0.9706070287539936
F1-Score : 0.9704348792479691
Sensitivity: 0.9558511988245508
Specificity: 0.9558511988245508

Training Size : 80%
Training Shape: (4172, 200, 200, 1)
Testing Shape : (1044, 200, 200, 1)
/usr/local/lib/python3.12/dist-packag
super().__init__(activity_regulariz

CNN Results (80-19 Split)
Accuracy : 0.9703065134099617
F1-Score : 0.97063076978051
Sensitivity: 0.9726977227265734
Specificity: 0.9726977227265734

Training Data Shape : (1043, 200, 200, 1)

Testing Data Shape : (4173, 200, 200, 1)

20-80 Split

CNN Results

Training Accuracy : 0.9856184124946594

Testing Accuracy : 0.9568655490875244

Training Data Shape : (2086, 200, 200, 1)

Testing Data Shape : (3130, 200, 200, 1)

40-60 Split

CNN Results

Training Accuracy : 0.9961649179458618

Testing Accuracy : 0.9670926332473755

Training Data Shape : (3129, 200, 200, 1)

Testing Data Shape : (2087, 200, 200, 1)

60-40 Split

CNN Results

Training Accuracy : 0.9964845180511475

Testing Accuracy : 0.9726880788803101

Training Shape : (4172, 200, 200, 1)

Testing Shape : (1044, 200, 200, 1)

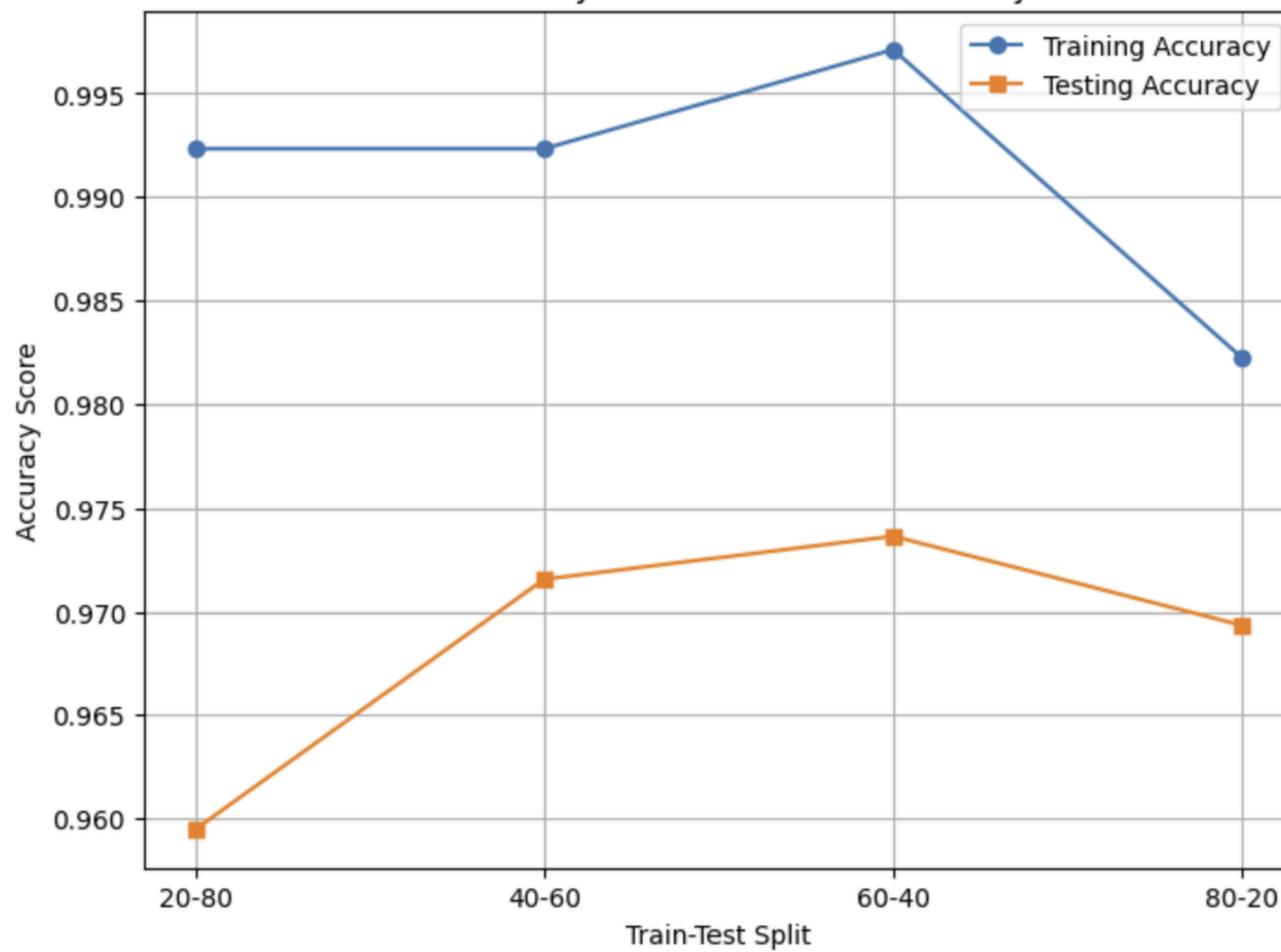
CNN Results

80-20 Split

Training Accuracy : 0.9952061176300049

Testing Accuracy : 0.967432975769043

Chest X-Ray CNN: Train vs Test Accuracy



TUMOR- TRANSFER LEARNING

Model: "sequential"

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 128)	163,968
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 4)	516

Total params: 2,422,468 (9.24 MB)

Trainable params: 164,484 (642.52 KB)

Non-trainable params: 2,257,984 (8.61 MB)

RESULTS

Training Accuracy: 0.9224738478660583
Testing Accuracy: 0.8763065934181213
F1-Score: 0.875891654982693
Sensitivity: 0.86253970666691
Specificity: 0.957063458130005

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.77	0.85	79
1	0.95	0.93	0.94	166
2	0.87	0.94	0.90	165
3	0.80	0.80	0.80	164
accuracy			0.88	574
macro avg	0.89	0.86	0.87	574
weighted avg	0.88	0.88	0.88	574

Confusion Matrix:

[[61	2	2	14]
[0	155	1	10]
[0	0	155	10]
[4	7	21	132]]
